

Maximum Likelihood Estimation

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The Likelihood Function

Let $X_1, \dots, X_n$ be an iid sample with pdf $f(x_i; \theta)$, where θ is a $(k \times 1)$ vector of parameters that characterize $f(x_i; \theta)$.

Example: Let $X_i \sim N(\mu, \sigma^2)$ then

$$\begin{aligned} f(x_i; \theta) &= (2\pi\sigma^2)^{-1/2} \exp\left(-\frac{1}{2\sigma^2}(x - \mu)^2\right) \\ \theta &= (\mu, \sigma^2)' \end{aligned}$$

The Likelihood Function

The *joint density* of the sample is, by independence, equal to the product of the marginal densities

$$f(x_1, \dots, x_n; \theta) = f(x_1; \theta) \cdots f(x_n; \theta) = \prod_{i=1}^n f(x_i; \theta).$$

The joint density is an n dimensional function of the data $x_1, \dots, x_n$ given the parameter vector θ and satisfies

$$\begin{aligned} f(x_1, \dots, x_n; \theta) &\geq 0 \\ \int \cdots \int f(x_1, \dots, x_n; \theta) dx_1 \cdots dx_n &= 1. \end{aligned}$$

The Likelihood Function

The *likelihood function* is defined as the joint density treated as a function of the parameters θ :

$$L(\theta|x_1, \dots, x_n) = f(x_1, \dots, x_n; \theta) = \prod_{i=1}^n f(x_i; \theta).$$

Notice that the likelihood function is a k dimensional function of θ given the data $x_1, \dots, x_n$.

Remarks

- For discrete variable, the likelihood function can also be defined as the joint probability mass function

Remarks

It is important to keep in mind that the likelihood function, being a function of θ and not the data, is not a proper pdf. It is always positive but

$$\int \cdots \int L(\theta|x_1, \dots, x_n) d\theta_1 \cdots d\theta_k \neq 1.$$

To simplify notation, let the vector $\mathbf{x} = (x_1, \dots, x_n)$ denote the observed sample. Then the joint pdf and likelihood function may be expressed as $f(\mathbf{x}; \theta)$ and $L(\theta|\mathbf{x})$, respectively.

Example 1. Bernoulli Sampling

Let $X_i \sim \text{Bernoulli}(\theta)$. That is,

$$\begin{aligned} X_i &= 1 \text{ with probability } \theta \\ X_i &= 0 \text{ with probability } 1 - \theta \end{aligned}$$

The pdf for X_i is

$$f(x_i; \theta) = \theta^{x_i}(1 - \theta)^{1-x_i}, \quad x_i = 0, 1$$

Let $X_1, \dots, X_n$ be an iid sample with $X_i \sim \text{Bernoulli}(\theta)$. The joint density / likelihood function is given by

$$f(\mathbf{x}; \theta) = L(\theta|\mathbf{x}) = \prod_{i=1}^n \theta^{x_i}(1 - \theta)^{1-x_i} = \theta^{\sum_{i=1}^n x_i} (1 - \theta)^{n - \sum_{i=1}^n x_i}$$

Since X_i is a discrete random variable

$$f(\mathbf{x}; \theta) = \Pr(X_1 = x_1, \dots, X_n = x_n)$$

Example 2. Normal Sampling

Let $X_1, \dots, X_n$ be an iid sample with $X_i \sim N(\mu, \sigma^2)$. The pdf for X_i is

$$f(x_i; \theta) = (2\pi\sigma^2)^{-1/2} \exp\left(-\frac{1}{2\sigma^2}(x_i - \mu)^2\right),$$

$$\theta = (\mu, \sigma^2)'$$

$$-\infty < \mu < \infty, \sigma^2 > 0, -\infty < x_i < \infty$$

The likelihood function is given by

$$\begin{aligned} L(\theta|\mathbf{x}) &= \prod_{i=1}^n (2\pi\sigma^2)^{-1/2} \exp\left(-\frac{1}{2\sigma^2}(x_i - \mu)^2\right) \\ &= (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2\right) \end{aligned}$$

The Maximum Likelihood Estimation (MLE)

Suppose we have a random sample from the pdf $f(x_i; \theta)$ and we are interested in estimating θ .

The maximum likelihood estimator, denoted $\hat{\theta}_{mle}$, is the value of θ that maximizes $L(\theta|\mathbf{x})$. That is,

$$\hat{\theta}_{mle} = \arg \max_{\theta} L(\theta|\mathbf{x})$$

Alternatively, we say that $\hat{\theta}_{mle}$ solves

$$\max_{\theta} L(\theta|\mathbf{x})$$

Why MLE?

“The event with largest probability is most likely to happen”

Remark

It is often quite difficult to directly maximize $L(\theta|\mathbf{x})$. It is usually much easier to maximize the log-likelihood function $\ln L(\theta|\mathbf{x})$. Since $\ln(\cdot)$ is a monotonic function

$$\hat{\theta}_{mle} = \arg \max_{\theta} \ln L(\theta|\mathbf{x})$$

With random sampling, the log-likelihood has the particularly simple form

$$\ln L(\theta|\mathbf{x}) = \ln \left(\prod_{i=1}^n f(x_i; \theta) \right) = \sum_{i=1}^n \ln f(x_i; \theta)$$

Example 3. Bernoulli example continued

Given the likelihood function

$$L(\theta|\mathbf{x}) = \theta^{\sum_{i=1}^n x_i} (1 - \theta)^{n - \sum_{i=1}^n x_i},$$

the log-likelihood is

$$\begin{aligned} \ln L(\theta|\mathbf{x}) &= \ln \left(\theta^{\sum_{i=1}^n x_i} (1 - \theta)^{n - \sum_{i=1}^n x_i} \right) \\ &= \left(\sum_{i=1}^n x_i \right) \ln(\theta) + \left(n - \sum_{i=1}^n x_i \right) \ln(1 - \theta) \end{aligned}$$

Example 4. Normal example continued

Given the likelihood function

$$\ln L(\theta|\mathbf{x}) = (2\pi\sigma^2)^{-n/2} \exp \left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 \right)$$

the log-likelihood is

$$\ln L(\theta|\mathbf{x}) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2.$$

Review of Optimization

Example: finding the minimum of a univariate function

$$y = f(x) = x^2$$
$$\min_x y = f(x)$$

First order conditions for a minimum

$$0 = \frac{df(x)}{dx} = \frac{d}{dx} 2x = 2 \cdot x$$
$$\Rightarrow x = 0$$

Second order conditions for a minimum

$$0 < \frac{d^2 f(x)}{dx^2} = \frac{d}{dx} 2 \cdot x = 2$$

Review of Optimization

Example: Finding the minimum of a bivariate function

$$y = f(x, z) = x^2 + z^2$$

$$\min_{x, z} y = f(x, z)$$

First order conditions for a minimum

$$0 = \frac{\partial f(x, z)}{\partial x} = \frac{\partial}{\partial x} (x^2 + z^2) = 2 \cdot x$$

$$0 = \frac{\partial f(x, z)}{\partial z} = \frac{\partial}{\partial z} (x^2 + z^2) = 2 \cdot z$$

$$\Rightarrow x = 0, z = 0$$

Hessian matrix

$$H(x, z) = \frac{\partial^2 f(x, z)}{\partial x \partial z} = \begin{bmatrix} \frac{\partial^2 f(x, z)}{\partial x^2} & \frac{\partial^2 f(x, z)}{\partial x \partial z} \\ \frac{\partial^2 f(x, z)}{\partial z \partial x} & \frac{\partial^2 f(x, z)}{\partial z^2} \end{bmatrix}$$

Example 5 Bernoulli example continued

To find the MLE for θ , we maximize the log-likelihood function

$$\ln L(\theta|\mathbf{x}) = \sum_{i=1}^n x_i \ln(\theta) + \left(n - \sum_{i=1}^n x_i \right) \ln(1 - \theta)$$

The derivative of the log-likelihood is

$$\frac{\partial \ln L(\theta|\mathbf{x})}{\partial \theta} = \frac{1}{\theta} \sum_{i=1}^n x_i - \frac{1}{1 - \theta} \left(n - \sum_{i=1}^n x_i \right) = 0$$

The MLE satisfies $\frac{\partial \ln L(\theta|\mathbf{x})}{\partial \theta} = 0$ and solving for θ gives

$$\hat{\theta}_{mle} = \frac{1}{n} \sum_{i=1}^n x_i.$$

Example 6 Normal example continued

To find the MLE for $\theta = (\mu, \sigma^2)'$, we maximize the log-likelihood function

$$\ln L(\theta|\mathbf{x}) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2.$$

The derivative of the log-likelihood is a (2×1) vector given by

$$\frac{\partial \ln L(\theta|\mathbf{x})}{\partial \theta} = \begin{pmatrix} \frac{\partial \ln L(\theta|\mathbf{x})}{\partial \mu} \\ \frac{\partial \ln L(\theta|\mathbf{x})}{\partial \sigma^2} \end{pmatrix}$$

Example 6 Normal example continued

where

$$\begin{aligned}\frac{\partial \ln L(\theta|\mathbf{x})}{\partial \mu} &= \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu) \\ \frac{\partial \ln L(\theta|\mathbf{x})}{\partial \sigma^2} &= -\frac{n}{2}(\sigma^2)^{-1} + \frac{1}{2}(\sigma^2)^{-2} \sum_{i=1}^n (x_i - \mu)^2\end{aligned}$$

Solving $\frac{\partial \ln L(\theta|\mathbf{x})}{\partial \theta} = 0$ gives the *normal equations*

$$\begin{aligned}\frac{\partial \ln L(\hat{\theta}_{mle}|\mathbf{x})}{\partial \mu} &= \frac{1}{\hat{\sigma}_{mle}^2} \sum_{i=1}^n (x_i - \hat{\mu}_{mle}) = 0 \\ \frac{\partial \ln L(\hat{\theta}_{mle}|\mathbf{x})}{\partial \sigma^2} &= -\frac{n}{2}(\hat{\sigma}_{mle}^2)^{-1} \\ &+ \frac{1}{2}(\hat{\sigma}_{mle}^2)^{-2} \sum_{i=1}^n (x_i - \hat{\mu}_{mle})^2 = 0\end{aligned}$$

Example 6 Normal example continued

Solving the first equation for $\hat{\mu}_{mle}$ gives

$$\hat{\mu}_{mle} = \frac{1}{n} \sum_{i=1}^n x_i = \bar{x}$$

Solving the second equation for $\hat{\sigma}_{mle}^2$ gives

$$\hat{\sigma}_{mle}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \hat{\mu}_{mle})^2.$$

Notice that $\hat{\sigma}_{mle}^2$ is not equal to the sample variance.

Invariance Property of MLE

One of the attractive features of the method of maximum likelihood is its invariance to one-to-one transformations of the parameters of the log-likelihood.

That is, if $\hat{\theta}_{mle}$ is the MLE of θ and $\alpha = h(\theta)$ is a one-to-one function of θ then $\hat{\alpha}_{mle} = h(\hat{\theta}_{mle})$ is the mle for α .

Example 7 Normal Model Continued

The log-likelihood is parameterized in terms of μ and σ^2 and

$$\begin{aligned}\hat{\mu}_{mle} &= \bar{x} \\ \hat{\sigma}_{mle}^2 &= \frac{1}{n} \sum_{i=1}^n (x_i - \mu_{mle})^2\end{aligned}$$

Suppose we are interested in the MLE for

$$\sigma = h(\sigma^2) = (\sigma^2)^{1/2}$$

which is a one-to-one function for $\sigma^2 > 0$.

The invariance property says that

$$\hat{\sigma}_{mle} = (\hat{\sigma}_{mle}^2)^{1/2} = \left(\frac{1}{n} \sum_{i=1}^n (x_i - \hat{\mu}_{mle})^2 \right)^{1/2}$$

Exercise:

MLE for Uniform (a, b)